PACK NODE (A19): MODULE CURRENT OVER THE DOCK PINS, PRE-CHARGE, GAUGE SHUNT, MEZZANINE FEED
CHARGER BQ25792 (SHORE_12V, ILIM_HIZ 3 A, JEITA ON THE MODULE THERMISTOR) + DOCK SIGNAL PINS
GAUGE BQ34Z100-G1 (0x55)
RAIL M1: F3 + TPS61288L 5.05 V (LOGIC, HUB, PCB-B M1)
RAIL M2: F4 + TPS61288L 5.05 V (PCB-B M2)
RAIL PI: F5 + TPS61288L 5.1 V 5 A
MAIN POWER CONTROL LTC2954 + HEATING PAD SWITCH TPS259571
3.3 V BUCK TPS563201 FROM M1, TEST POINTS, FLAGS
J_CP1 CELL+ 9 A spring pin, CELL+ (Mill-Max 0858 class, dock block)
J_CP2 CELL+ 9 A spring pin, CELL+ (Mill-Max 0858 class, dock block)
J_CP3 CELL+ 9 A spring pin, CELL+ (Mill-Max 0858 class, dock block)
J_CP4 CELL+ 9 A spring pin, CELL+ (Mill-Max 0858 class, dock block)
J_CN1 CELL_N 9 A spring pin, module return CELL_N (Mill-Max 0858 class, dock block)
J_CN2 CELL_N 9 A spring pin, module return CELL_N (Mill-Max 0858 class, dock block)
J_CN3 CELL_N 9 A spring pin, module return CELL_N (Mill-Max 0858 class, dock block)
J_CN4 CELL_N 9 A spring pin, module return CELL_N (Mill-Max 0858 class, dock block)
J_PRE PRECHG pre-charge pin, longer, mates first (32.24 AX)
R51 10R 2W 2512
R52 3 mOhm 1% 3 W 2512 shunt (RALEC LR2512-23R003F4): gauge SRP/SRN Kelvin (32.24 AV)
F2 10 A mini blade (Keystone 3568 holder): pack node to the 8 V boost feed
MEZZ_CELL J_MEZZ_PWR1 mezzanine 8 V boost feed (JST-VH): cell node through F2 (R15)
C7 10u
C50 100u 10V
U20 BQ25792 1S charger from SHORE_12V (3 A set over ILIM_HIZ, 750 kHz, I2C 0x6B)
CHG_STAT SHORE_12V SHORE_12V BTST1 REGN SHORE_12V SHORE_12V QON SCL SDA TS_CHG ILIM_HIZ CELL_SENSE_P BTST2 PROG CHG_INT CELL+ CELL+ SDRV SYS_CHG SW2_CHG SW1_CHG PMID
L5 2.2uH XAL4030-222MEB (Isat 7.4 A) between SW1 and SW2 (750 kHz, PROG 4.7k)
C51 10u 25V 1210
C52 10u 25V 1210
C53 10u 25V 1210
C54 10u 25V 1210
C55 10u 25V 1210
C56 10u 25V 1210
C57 10u 25V 1210
C58 10u 25V 1210
C59 10u 25V 1210
C60 10u 25V 1210
C61 10u
C62 10u
C63 47n
C64 47n
C65 4.7u
C66 1n
C67 100n
R53 4.7k 1% (PROG: 1S, 750 kHz)
R54 16.5k 1%
R55 34.8k 1%
R56 5.23k 1% (RT1 JEITA)
R57 30.1k 1% (RT2 JEITA)
R58 100k
R59 10k
R60 10k
J_DOCK spring pins to the dock block (2x6, Preci-Dip 813-S1-012-10-016101, underside): 1-4 SHORE_12V, 5-7 GND, 8 SHORE_INHIBIT, 9 module thermistor (103AT to GND), 10 GND, 11 Kelvin cell sense +, 12 spare
SHORE_12V SHORE_12V SHORE_12V SHORE_12V SHORE_INHIBIT TS_CHG CELL_SENSE_P DOCK_SPARE
U21 BQ34Z100-G1 gauge (I2C 0x55, SCALED for 42 Ah, 3 mOhm shunt)
GAUGE_ALERT CELL_SENSE_P CELL+ CELL+ REG25 CELL_N CELL_N TS_GAUGE SCL SDA
C68 100n
C69 1u
RT1 10k NTC 0603 1% (gauge temperature, beside the dock pins; the JEITA sensor is the module's own 103AT-2)
F3 10 A mini blade (Keystone 3568 holder): pack node to the M1 converter
U22 TPS61288L boost +5V_M1 (15 A switch, 500 kHz, no MODE pin)
FB1 COMP1 SW1 +5V_M1 BOOST_EN BOOST1_IN BST1 SW1 VCC1
L2 2.2uH XAL1010-222MED (Isat 34 A, 10 mm)
C38 100n 25V
C39 2.2u
R44 8.87k
C40 3.3n
C41 27p
R47 102k 1%
R48 13.7k 1%
C42 22u 10V X7R 1210
C43 22u 10V X7R 1210
C44 100n
C45 22u 10V X7R 1210
C46 22u 10V X7R 1210
C47 22u 10V X7R 1210
C48 22u 10V X7R 1210
C49 22u 10V X7R 1210
C70 22u 10V X7R 1210
C95 100u 10V
J_5V_M1 +5V_M1 to PCB-B J_5V_M1 (JST-VH, 18 AWG): + −
F4 10 A mini blade (Keystone 3568 holder): pack node to the M2 converter
U23 TPS61288L boost +5V_M2 (15 A switch, 500 kHz, no MODE pin)
FB2 COMP2 SW2 +5V_M2 BOOST_EN BOOST2_IN BST2 SW2 VCC2
L3 2.2uH XAL1010-222MED (Isat 34 A, 10 mm)
C71 100n 25V
C72 2.2u
R61 8.87k
C73 3.3n
C74 27p
R62 102k 1%
R63 13.7k 1%
C75 22u 10V X7R 1210
C76 22u 10V X7R 1210
C77 100n
C78 22u 10V X7R 1210
C79 22u 10V X7R 1210
C80 22u 10V X7R 1210
C81 22u 10V X7R 1210
C82 22u 10V X7R 1210
C83 22u 10V X7R 1210
C96 100u 10V
J_5V_M2 +5V_M2 to PCB-B J_5V_M2 (JST-VH, 18 AWG): + −
F5 15 A mini blade (Keystone 3568 holder): pack node to the PI converter
U24 TPS61288L boost +5V_PI (15 A switch, 500 kHz, no MODE pin)
FB3 COMP3 SW3 +5V_PI BOOST_EN BOOST3_IN BST3 SW3 VCC3
L4 2.2uH XAL1010-222MED (Isat 34 A, 10 mm)
C84 100n 25V
C85 2.2u
R64 8.87k
C86 3.3n
C87 27p
R65 75k 1%
R66 10k 1%
C88 22u 10V X7R 1210
C89 22u 10V X7R 1210
C90 100n
C91 22u 10V X7R 1210
C92 22u 10V X7R 1210
C93 22u 10V X7R 1210
C94 22u 10V X7R 1210
C97 100u 10V
J_5V_PI +5V_PI 5.1 V 5 A to PCB-B J_5V_PI (JST-VH, 18 AWG): + −
U25 LTC2954CTS8-1 push-button on/off controller
CELL+ MAIN_PB PI_SHDN_REQ BOOST_EN KILL
C98 1u
R67 100k
R68 100k
R69 100k
Q5 2N7002: PI GPIO high = pull KILL low = power off
PI_KILL
R70 100k
J_MAINSW MAIN button lead from the panel (XH2.5): PB, GND
F6 2.5A hold 30V 1812
U26 TPS259571DSGR eFuse 12 V 2.0 A for the heating pad
HEAT_DVDT HEAT_EN HEAT_IN HEAT_IN HEAT_OUT HEAT_FLT HEAT_ILM
C99 10n
R71 1.02k 1% (2.0 A)
R72 10k
R73 100k
C100 100n
J_HEAT 12 V heating pad on the battery module (XH2.5): + −
U5 TPS563201 3.3 V buck from M1
VIN EN GND SW VBST VFB
SW33 BST33 FB33
L6 3.3uH XAL4020-332MEB
C13 100n
C14 10u
C15 22u 10V X7R 1210
C101 22u 10V X7R 1210
R74 33.2k 1%
R75 10k 1%
D2 SML-LX0
R18 1k
LED_PWR_A
TP1 +5V_M1
TP2 GND
TP3 +3V3
TP4 CELL_N
TP5 CELL+
TP6 SHORE_INHIBIT
TP7 GAUGE_ALERT
TP8 MEZZ_SPARE1
TP9 TX_INHIBIT_n
TP10 SHORE_12V
TP11 CHG_INT
TP12 +5V_M2
TP13 +5V_PI
TP14 DOCK_SPARE
TP15 BOOST_EN
TP16 CELL_SENSE_P
#FLG01 PWR_FLAG CELL+
#FLG02 PWR_FLAG CELL_N
#FLG03 PWR_FLAG GND
#FLG04 PWR_FLAG +3V3
#FLG05 PWR_FLAG +5V_M1
#FLG06 PWR_FLAG +5V_M2
#FLG07 PWR_FLAG +5V_PI
#FLG08 PWR_FLAG 5V_WIFI
#FLG09 PWR_FLAG 5V_GPS
#FLG10 PWR_FLAG 5V_CODEC
#FLG11 PWR_FLAG 5V_UART
#FLG12 PWR_FLAG 5V_WALL
#FLG13 PWR_FLAG BOOST1_IN
#FLG14 PWR_FLAG BOOST2_IN
#FLG15 PWR_FLAG BOOST3_IN
#FLG16 PWR_FLAG SHORE_12V
#FLG17 PWR_FLAG HEAT_IN
HEAT_OUT REGN SYS_CHG PMID REG25

MESHSAT-709 / MESHSAT-789. A19 (4 Sep 2026 rulings, appendix 32.13 to 32.25): the kit UPS. BQ2
Phase A2 schematic (A4: 16.3 fixes), generated by tools/gen_sch_a.py. Netlist style: every pin carries

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File: pcb-a-power.kicad_sch

Title: MeshSat Field Kit carrier – PCB-A POWER + I/O

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